As long as we have a *complete* description of a system, all the variables of the system are on the same footing; none of them acts as a time variable. That is to say: none is correlated to irreversible phenomena. But as soon as we describe the system by means of averages of many variables, we have a preferred variable that functions like common time. A time along which heat is dissipated. The time of our everyday experience.

Hence time is not a fundamental constituent of the world, but it appears because the world is immense, and we are small systems within the world, interacting only with macroscopic variables that average among innumerable small, microscopic variables. We, in our everyday lives, never see a single elementary particle, or a single quantum of space. We see stones, mountains, the faces of our friends – and each of these things we see is formed by myriads of elementary components. We are always correlated with averages. Averages behave like averages: they disperse heat and, intrinsically, generate time.

The difficulty of grasping this idea comes from the fact that it is hard for us to think of a world without time, and of time emerging in an approximate manner. We are too used to thinking of reality as existing in time. We are beings who live in time: we dwell in time, and are nourished by it. We are an effect of this temporality, produced by average values of microscopic variables. But the limitations of our intuitions should not mislead us. Understanding the world better often entails going against intuition. If this were not the case, understanding would be easy.

Time is an effect of our overlooking of the physical microstates of things. Time is information we don't have.

Time is our ignorance.

Reality and information

Why does information play such a central role as this? Perhaps because we must not confuse what we know about a system with the absolute state of the same system. What we know is something concerning the relation between the system and ourselves. Knowledge is intrinsically relational; it depends just as much on its object as upon its subject. The notion of the 'state' of a system refers, explicitly or implicitly, to another system. Classical mechanics misled us into thinking that we could do without taking account of this simple truth, and that we could access, at least in theory, a vision of reality entirely independent of the observer. But the development of physics has shown that, at the end of the day, this is impossible.

Careful: when I say that we 'have information' about the temperature of cup of tea, or we 'don't have information' about the velocity of every single molecule, I am not saying something about mental states, or abstract ideas. I